Joseph Walton

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Professional Summary

Data scientist and machine learning engineer with over 10 years of experience delivering data-driven solutions across government, intellectual property, financial services and defence. Strong background in NLP, text analytics and unstructured data, including multilingual language models, semantic search, recommendation systems, RAG and generative AI. Experienced across the full machine learning life cycle, from discovery and experimentation through evaluation, deployment, monitoring and continuous improvement. Independent technical leader who works effectively with business and engineering stakeholders, challenges assumptions with evidence, and communicates complex findings, risks and recommendations clearly.

Technical Skills

- **Programming & Data:** Python, SQL, Java, Pandas, Polars, NumPy, Apache Spark
- **NLP & Generative AI:** Hugging Face, BERT, FastText, Azure AI Foundry LangChain, LangGraph, RAG, vector search
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch
- **MLOps & Engineering:** MLflow, Azure ML Pipelines, Vertex AI Pipelines, CI/CD
- **Cloud Platforms:** Microsoft Azure, Google Cloud Platform, AWS

Professional Experience

IBM

Lead Data Scientist
June 2025 - Present
Contract role delivering AI solutions for central government.

- Rescued underperforming AI projects by challenging proposed approaches and introducing evidence-based evaluation and observability practices.
- Designed and delivered an agent-based data processing solution using Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.
- Re-engineered a data matching algorithm reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.

Tech Stack: Python, Polars, Azure AI Foundry, Azure Functions, LangGraph

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Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Developed and operationalised NLP solutions to improve trade mark application outcomes in a government environment.

- Set the product roadmap for AI applications in the intellectual property domain, evaluating opportunities, risks and practical delivery constraints.
- Communicated the IPO's approach to AI implementation to technical and non-technical stakeholders across multiple international IP offices.
- Contributed to cross-government AI policy and best practices through the government AI working group.
- Developed generative AI applications within a service-oriented architecture using LangChain, Azure OpenAI and Azure Functions.
- Built a RAG agent that used unstructured text to help novice users identify relevant Goods and Services terms.
- Conducted red-team experiments to evaluate generative AI robustness and identify improvements against adversarial attacks.
- Implemented and tuned Elasticsearch HNSW vector search, balancing throughput and recall through systematic evaluation.
- Used relevance metrics and experimental evidence to refine search and recommendation systems, improving relevance by 20%.
- Implemented an end-to-end MLOps strategy covering deployment across multiple environments, reducing model deployment time to under one day.
- Architected and built ETL pipelines in Polars to transform complex datasets, replacing Azure Databricks workloads, reducing compute costs by 50% and enabling zero-downtime deployments.

Tech Stack: Python, Azure OpenAI, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

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Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led cross-functional delivery of production machine learning solutions using large-scale text and image data.

- Led the design and implementation of an MLOps framework on Google Cloud Platform, orchestrating training, evaluation, approval and deployment with Vertex AI Pipelines and reducing time to market to under one day.
- Built Spark data pipelines to clean and transform large-scale text and image datasets for model training, batch inference and recommendation workloads.
- Developed and deployed NLP and computer vision models for intellectual property protection, including a multilingual BERT model that detected infringements across more than 90 languages.
- Implemented a recommendation engine combining text and image similarity to improve IP violation detection.
- Mentored a software engineer through a transition into a machine learning role.

Tech Stack: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

Researched, evaluated and deployed machine learning models for intellectual property protection.

- Designed and built a production NLP system for predicting IP infringement in a secure, containerised environment, increasing enforced infringements by 10% per analyst across a workforce of approximately 200 analysts.

- Built and deployed an image similarity model that helped secure a major blue-chip client contract representing 10% of annual recurring revenue.

Tech Stack: Python, TensorFlow, FastText, Google Cloud Platform, MLflow, Docker

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Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

Delivered national-scale government data integration projects with a focus on secure data handling.

- Developed machine learning models and scalable data pipelines to link and process administrative datasets from multiple sources in support of national surveys.
- Worked in a cross-functional team on the UK Census 2021 programme, developing a highly scalable backend for the census data collection system.

Tech Stack: Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018

Seconded to Barclays to develop regional economic indicators from payments data.

- Applied statistical and analytical methods to financial data for novel statistical measures of economic activity.

This continued work from the winning ONS x Barclays Hackathon team.

Purple Secure Systems

Software Engineer

August 2014 - October 2016

Developed big data processing systems for government and defense clients.

Tech Stack: Java, Python, PySpark, AngularJS

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & Awards

Winner of ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates for Gross Value Added (GVA).